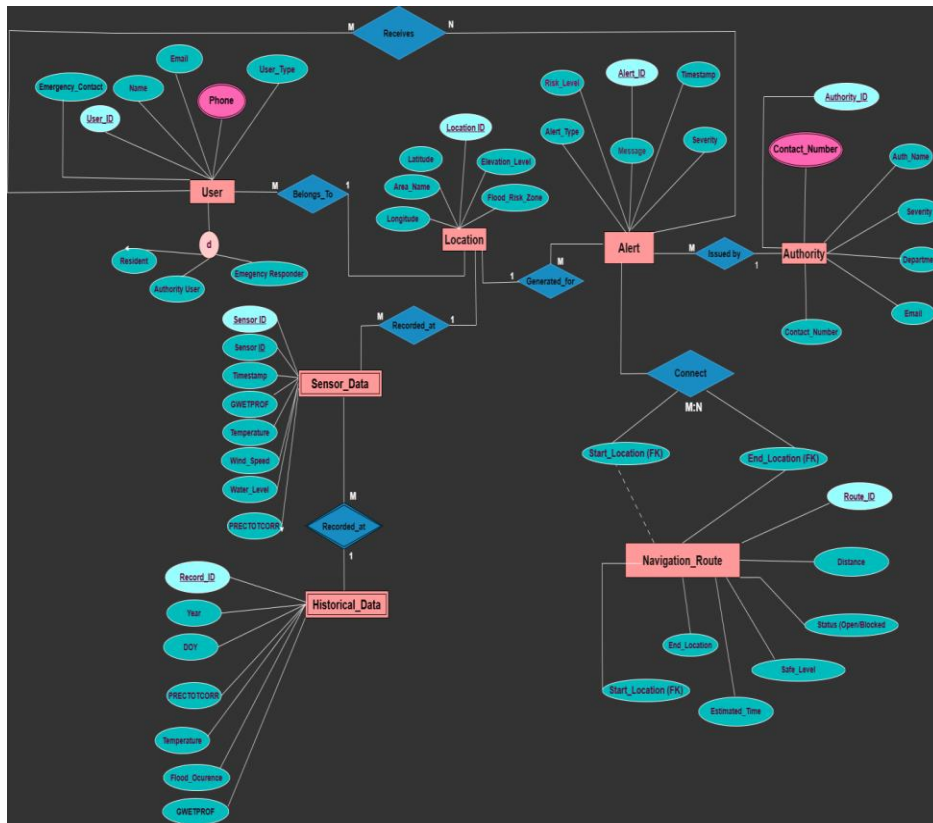


System Design & Entity-Relationship Modeling for AI-Based Flood Warning and Navigation System



Summary

This document outlines the data architecture for a real-time **AI-driven Flood Warning and Navigation System**. The system is designed to ingest environmental data from IoT sensors and historical records to provide early warnings and safe navigation routes for users and authorities.

2. Core Entities and Roles

- **User:** Stores profiles for Residents, Authorities, and Emergency Responders.
- **Sensor_Data:** Records environmental metrics such as water levels, wind speed, and precipitation.
- **Location:** Manages geospatial data, including risk zones, elevation levels, and coordinates.
- **Alert:** Handles automated warning generation based on risk levels (High, Medium, Low).
- **Navigation_Route:** Focuses on safety by calculating safe transit paths and estimated arrival times during floods.